



Operator Perception

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC
Management

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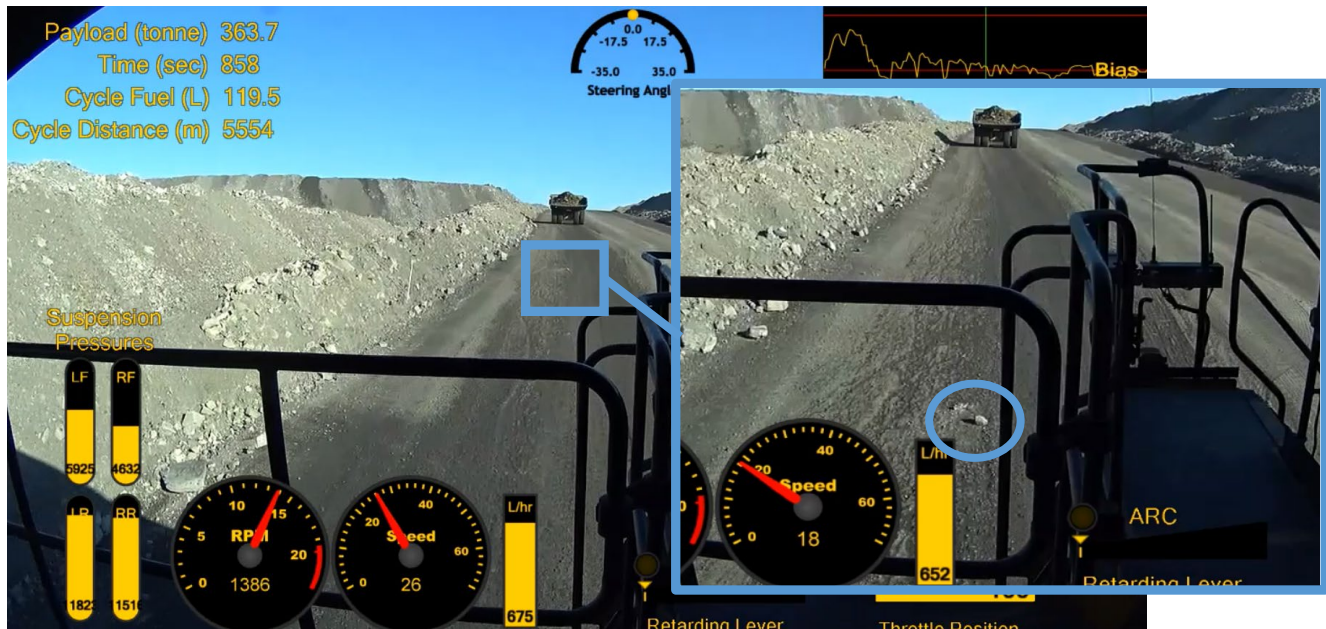
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February 2022
2.04-220218-1

1.0 Introduction

Tire expenses on large mining trucks is often a very significant portion 8-15% of the hourly operating costs of the hauling fleet. “A rock the size of a fist is large enough to cause damage to equipment tires” is a common statement that is used to help educate operators as to what size spillage and rocks should be avoided along the haul road.

When the operator returns to the cab of a mining truck however, a rock that could damage the tires can now appear extremely small and insignificant. The perception of the rocks spilt onto the haul road changes significantly and the operator may elect to drive over these sections without full recognition of the potential consequences.



2.0 Best Practice Description

To improve these operator perception issues Bulga Coal Mine site has initiated a tire awareness campaign utilizing Wiffleballs. These wiffleballs are highly visible and of an equivalent size to the rocks that the pit supervisors would like operators to avoid however if hit will not cause damage to the tires. Details as to how this campaign is managed is listed within the implementation steps below.



Wiffleball Campaign
Softball Diameter – 8.9 cm
Baseball Diameter – 7.6cm

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3.0 Implementation Steps

1. During Pre-shift meeting, pit supervisor raises the issue of damaging tires and quotes statistics on the costs to their site due to rock cuts.
2. The whiffle balls are passed around to operators as an example of the size of rocks that can cause damage.
3. During the shift, the pit-supervisor is to distribute the brightly colored whiffleballs within the load areas, along the haul roads and dump areas where spillage is often found.
4. Operators are then able to identify the whiffleball along the road and understand what a rock of that size would look like from the cab of a truck.
(The intent of the campaign is not to treat the whiffleballs as if they were a rock. Operators can elect to run over / straddle / drive around the balls during the shift. The intent is to ensure that they improve the understanding of what rocks should be avoided in future.)
5. Upon the completion of shift, a debrief is conducted to discuss how the operator perception campaign has affected operators and to hear their thoughts on the exercises.
6. Ongoing monitoring of tire average life will establish the value of this program on your site.



4.0 Benefits

There are many benefits that were observed throughout the campaign. In addition to the direct perception improvements and corresponding average tire life improvements, the whiffleballs were seen in many areas throughout the site. They were on display within the prestart and operator lunchrooms. Most managers had a whiffleball on their desks all highlighting their support for a common goal to improve tire life and reduce costs on site.

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		As Is	2% Tire Life Increase
Per Truck Estimates	Avg tire cost (\$)	\$ 30,000	\$ 30,000
	Avg tire life (hrs)	5000	5100
	Tires/Truck	6	6
	Operating Cost (\$/hr)	\$ 36.00	\$ 35.29
	Avg Operating hrs/Annum	6000	6000
	Truck Operating Costs (Tires)	\$ 216,000.00	\$ 211,764.71
Per Fleet Estimates	Truck fleet size	10	10
	Tire Cost/Hr (Fleet)	\$ 360.00	\$ 352.94
	Tire Cost/Year (Fleet)	\$ 2,160,000.00	\$ 2,117,647.06
	Annual Value	\$	42,353

5.0 Resources Required

The costs of the wiffleball campaign are relatively low. Bulk purchases of the wiffleballs will reduce the per unit cost and should be well under \$1 per ball. With the above projections and site history of rock related tire damage it is very easy to justify a campaign of this nature.

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

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